34 The potential difference (p.d.) across a fixed resistor is V. The power dissipated in the resistor is 5.0 W.

The p.d. across the resistor then changes to a new value. With the new p.d. the energy transferred to the resistor in a time of 2.25 s is 45.0 J.

What is the new p.d. across the resistor?

A $\frac{1}{2}V$ **B** $2V$ **C** $3V$ **D** $4V$